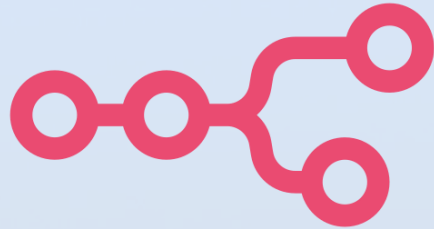


Introduction to AI Agent Tool



n8n

Rev: 2

By: Henry Chan

Date: Feb 2026

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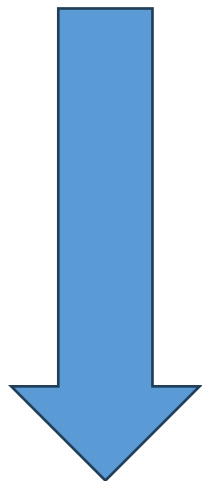
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Agenda

1. Introduction to AI Agent and n8n
2. Automation by Workflows
3. Important Ground Concepts

Introduction

The AI Evolution



Machine Learning

- Supervised and unsupervised learning
- Decision tree models
- Clustering....

Gen AI

- LLM (Large Language Model)
- Openai/ Gemini/ DeepSeek.....

AI Agent

- Understand the context (reasoning)
- Enable the user of tools automatically

Installation of n8n

Step 1: Download and install node.js

- Go to <https://nodejs.org/> and download the Windows installer MSI for Node.js (LTS recommended, version 20.19 – 24.x) from the official site and run it.
- After installation, open command prompt by typing “cmd” in Windows search box and enter.
- A prompt window will go out and you can verify the installation by typing:

```
C:\Users\NingNing>npm -v  
10.9.3  
  
C:\Users\NingNing>node -v  
v22.19.0
```

- n8n needs Node.js $\geq 20.19.x$ and $\leq 24.x$

Installation

There is a cloud version of n8n, but it is charged!

Step 2: Install n8n into your PC with npm

- npm* stands for “Node Package Manager” and is the world's largest software registry. It is installed with Node.js. The registry contains over 800,000 code packages. Open-source developers use *npm* to share software.
- Continue in the command prompt, type "**npm install -g n8n@1.122.5**" to install. It will take around 5 minutes.
- After finish, type "n8n" to run (if you are using PowerShell, you need to type "n8n.cmd start").
- Press "o" and the browser will open n8n web page.

```
C:\Users\NingNing>n8n
Initializing n8n process
n8n ready on ::, port 5678

There are deprecations related to your environment variables. Please take th
- DB_SQLITE_POOL_SIZE -> Running SQLite without a pool of read connections
e: https://docs.n8n.io/hosting/configuration/environment-variables/database/
- N8N_RUNNERS_ENABLED -> Running n8n without task runners is deprecated. Ta
NNERS_ENABLED=true` to enable task runners now and avoid potential issues in
/
- N8N_BLOCK_ENV_ACCESS_IN_NODE -> The default value of N8N_BLOCK_ENV_ACCESS
access environment variables from the Code Node or from expressions, please
g/configuration/environment-variables/security/

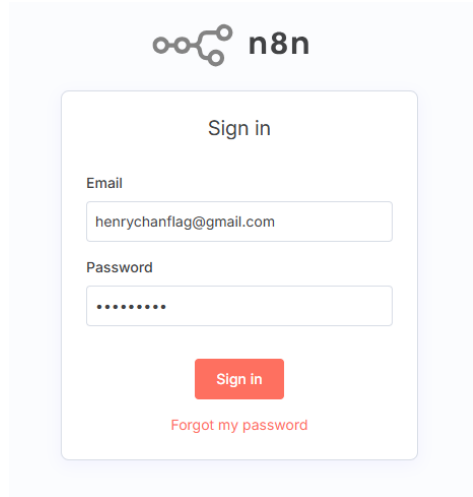
Initializing AuthRolesService...
AuthRolesService initialized successfully.
[license SDK] Skipping renewal on init: license cert is not initialized
Version: 1.109.2

Editor is now accessible via:
http://localhost:5678

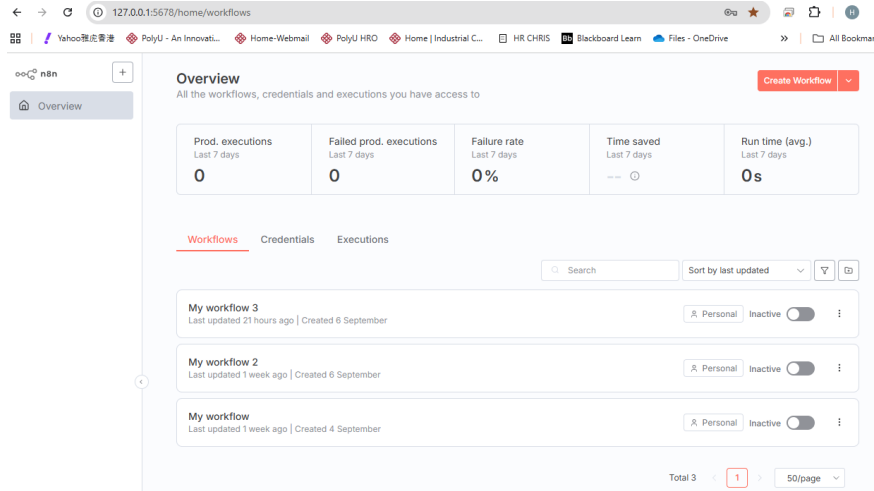
Press "o" to open in Browser.
```


Running n8n

- Like Node-RED, n8n uses web browser as the UI and coding console.
- Open the browser and type in the address <http://localhost:5678/> to get this console. n8n cloud version is charged while local version is free.
- Enter your email address as the owner of this n8n application.



The image shows the n8n sign-in interface. At the top is the n8n logo. Below it is a 'Sign in' heading. There are two input fields: 'Email' with the value 'henrychanflag@gmail.com' and 'Password' with masked characters. A red 'Sign in' button is at the bottom, with a link 'Forgot my password' below it.



The image shows the n8n Overview dashboard in a web browser. The address bar shows '127.0.0.1:5678/home/workflows'. The dashboard has a sidebar with 'Overview' selected. The main area shows an 'Overview' section with a table of statistics:

Prod. executions Last 7 days	Failed prod. executions Last 7 days	Failure rate Last 7 days	Time saved Last 7 days	Run time (avg.) Last 7 days
0	0	0%	---	0s

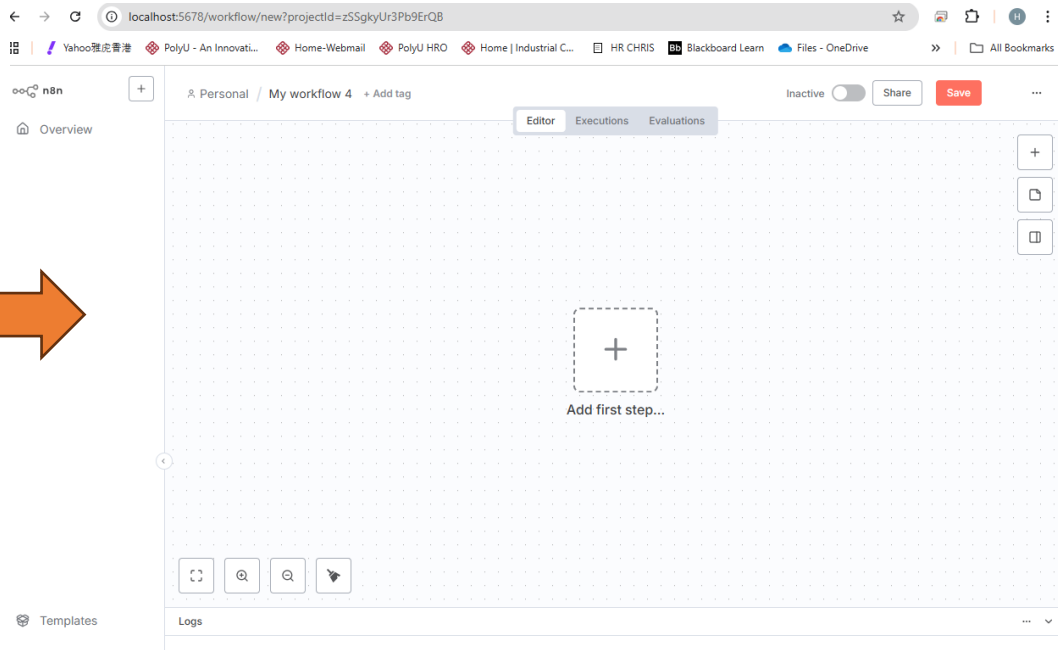
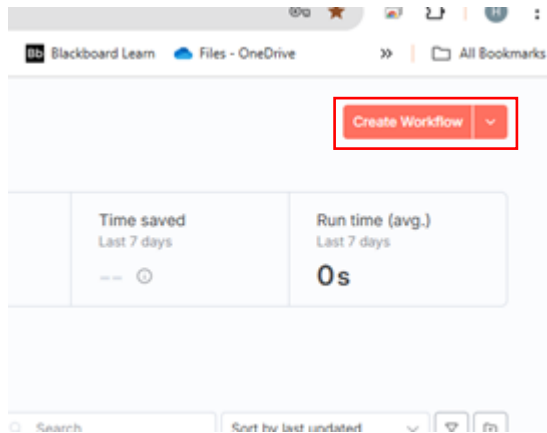
Below the statistics is a 'Workflows' section with tabs for 'Workflows', 'Credentials', and 'Executions'. It shows a list of workflows:

- My workflow 3**: Last updated 21 hours ago | Created 6 September. Status: Personal, Inactive.
- My workflow 2**: Last updated 1 week ago | Created 6 September. Status: Personal, Inactive.
- My workflow**: Last updated 1 week ago | Created 4 September. Status: Personal, Inactive.

At the bottom, it shows 'Total 3' workflows, with a pagination control set to '1' of '50/page'.

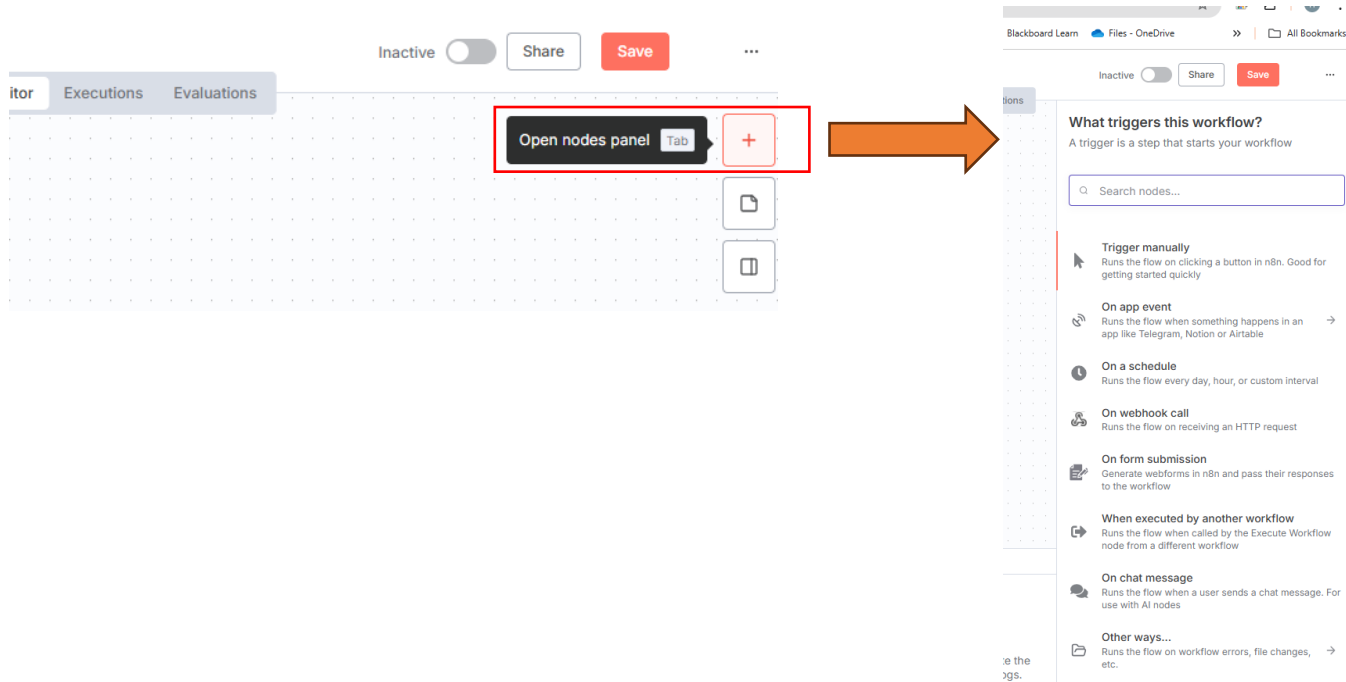
Creating your workflow

- In the *Overview* screen, select *Create Workflow*.
- Then it will go to the workflow workspace, and you are ready to create your workflow!



The Workflow Workspace

- Inside the workflow workspace, you can add nodes for your workflow:

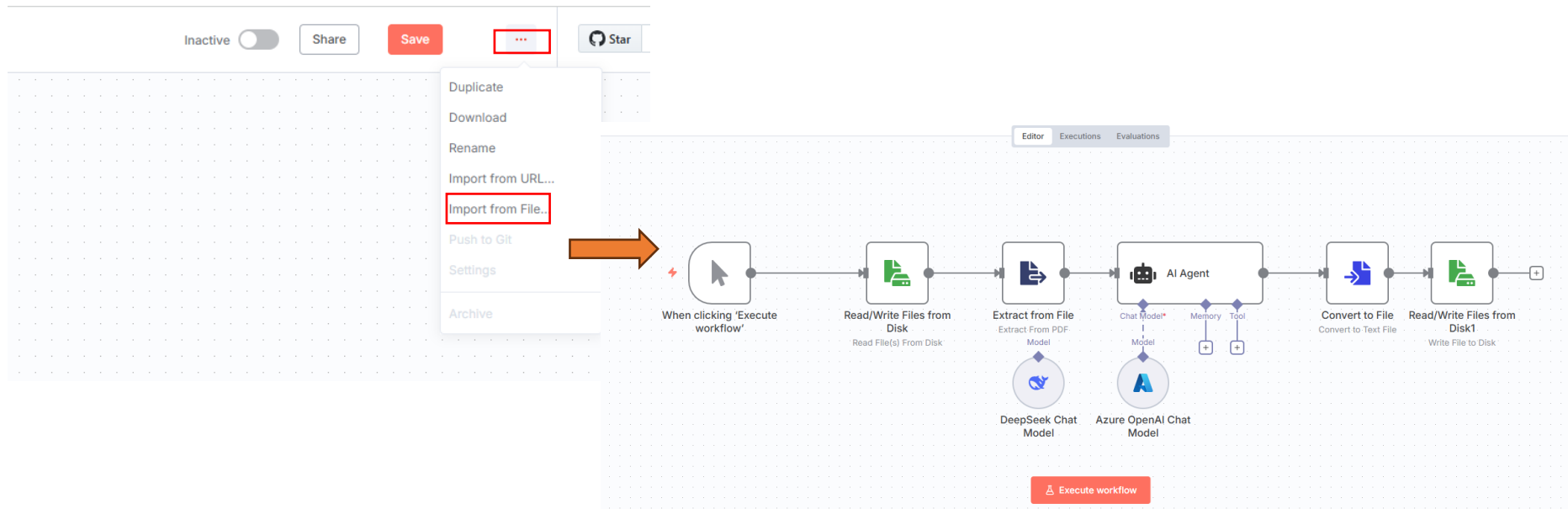




Automation by Workflows

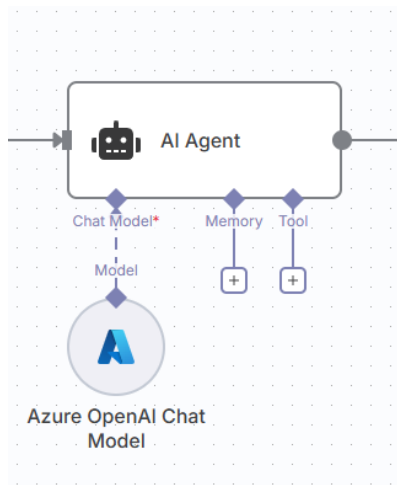
Your First Workflow

- Choose *import* to upload the flow "CV reading machine" to the workspace.



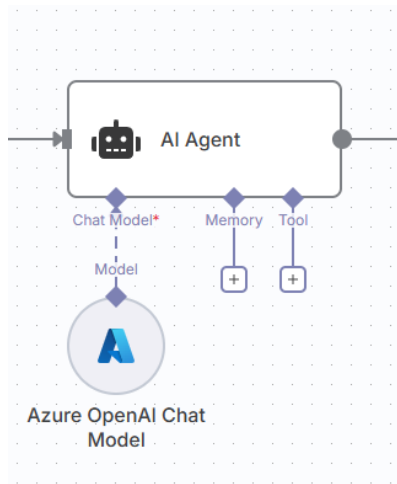
Example 1: CV Reading Machine

- This workflow helps us to read candidate's CV in pdf format, understand and analyze his education background and give a score (max. 10) based on his computer science and engineering academic strength.
- The most critical node is *AI Agent*, which uses LLM (Large Language Model) to do our task. Our first LLM deployed is *OpenAI ChatGPT*.



Example 1: CV Reading Machine

- This workflow helps us to read candidate's CV in pdf format, understand and analyze his education background and give a mark (max. 10) based on his computer science and engineering academic strength.
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The Core: AI Model

- You need to attach the AI model for the AI Agent to use. In this flow, we will try two kinds of AI Chat models: Azure OpenAI and DeepSeek
- You need to acquire these services by paying charges:
 - For OpenAI, you can get via Microsoft Azure Cloud Service (quite complicated).
 - For DeepSeek, it is much simpler. Just go to <https://www.deepseek.com/> and sign in an account. Then go to https://platform.deepseek.com/api_keys and pay for tokens. The cost is counted by tokens and now is very cheap. Below shows the price (in USD) of DeepSeek V3.2 api:

PRICING	1M INPUT TOKENS (CACHE HIT)	\$0.028
	1M INPUT TOKENS (CACHE MISS)	\$0.28
	1M OUTPUT TOKENS	\$0.42





AI Model – Create Credential

- Click the node to create a new credential with your custom name.
- Click *edit* icon to fill in the details as follows. The api key is shown in a separate file.

Azure OpenAI Chat Model

Parameters Settings Docs

Credential to connect with

Azure Open AI account

This node must be connected to an AI chain. [Insert one](#)

Model (Deployment) Name

gpt-4.1-Henry

Options

No properties

Add Option



Azure Open AI account
Azure Open AI

Connection

Need help filling out these fields? [Open docs](#)

Sharing

Details

API Key *

Resource Name *

Openai-Henry

API Version *

2025-03-01-preview

Endpoint

https://openai-henry.openai.azure.com/

Enterprise plan users can pull in credentials from external vaults. [More info](#)



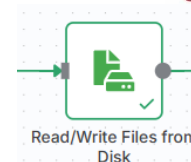
AI Model – Create Credential

- Click the node to create a new credential with your custom name.
- Click *edit* icon to fill in the details as follows. The api key is shown in a separate file.

- “Connection tested successfully” is shown. Now your AI models are ready to use!

Read/Write Files from/to Disk

- As the node name indicated, it reads or writes file content.
- Click the node and enter the parameters:



Read/Write F... Execute step

Parameters Settings Docs

Use this node to read and write files on the same computer running n8n. To handle files between different computers please use other nodes (e.g. FTP, HTTP Request, AWS).

Operation

Read File(s) From Disk

File(s) Selector

d:/cv sample.pdf

Supports patterns, learn more [here](#)

Read/Write Files from D... Execute step

Parameters Settings Docs

Use this node to read and write files on the same computer running n8n. To handle files between different computers please use other nodes (e.g. FTP, HTTP Request, AWS).

Operation

Write File to Disk

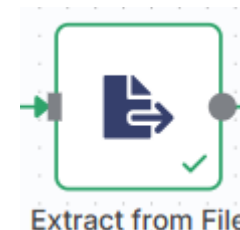
File Path and Name

d:/keypoints.doc

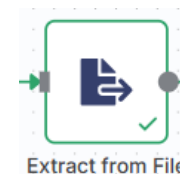
Input Binary Field

data

The name of the input binary field containing the file to be written



Note the slash "/"



Extract from File

- As the node name indicated, it extracts the contents from the file for the AI Agent to process.
- Click the node and enter the parameters:

 Extract from File

Parameters Settings

Operation

Extract From PDF

Input Binary Field

data

The name of the input binary field containing the file to be extracted

Options

No properties

Add option

- As the node name indicated, it converts the output from AI Agent into a file in a dedicated format.
- Click the node and enter the parameters:

.txt format

output from the AI Agent

The Main Role: AI Agent

- AI Agent is the key to perform your task using LLM.
- It uses LLM to understand your order and perform the task. Then output the answer in human language using LLM.
- Click the node and enter the parameters:

AI Agent

Execute step

Parameters Settings Docs

Tip: Get a feel for agents with our quick [tutorial](#) or see an [example](#) of how this node works

Source for Prompt (User Message)

Define below

Prompt (User Message)

fx

{{ \$json.text }}

ANUVA GOYAL D.O.B.: 1st October 2000 Gender: Female OBJECTIVE Energetic, innovative engineer...

Require Specific Output Format

Enable Fallback Model

Options

System Message

This is a resume of a candidate. Give a summary of his education and give a score at the end, a maximum of 10 if his education is related to engineering.

This is your order to the AI Agent;
That is the **prompt**.

Prompt Design – Prompt Engineering

- Prompt engineering is the practice of designing and refining prompts—questions or instructions.
- Key elements:
 - Instruction: The main task you want the AI to perform.
 - Context: Background information that helps the AI understand the scenario.
 - Examples: Sample inputs and outputs, especially in complex tasks (this technique is called **few-shot prompting**).

Natural Language Programming:

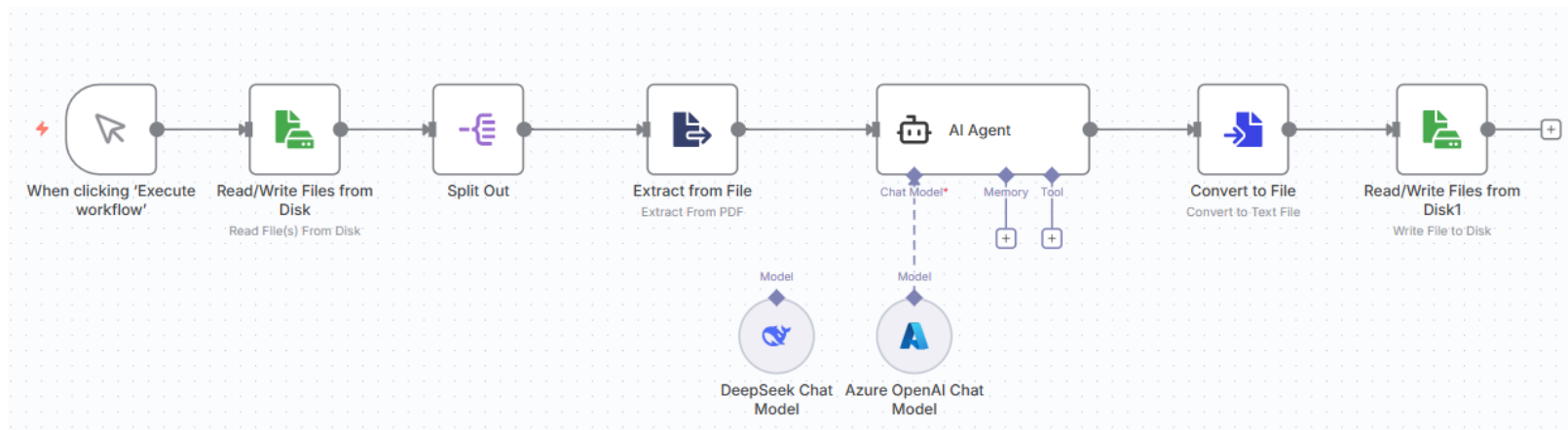
Managing human languages becomes more important than managing programming languages!

Your Turn Now – Exercise 1

- Using the original flow, modify it to generate some CVs of virtual candidates with similar or different backgrounds of the input file "cv sample.pdf".
- Use prompt to instruct AI Chat Model to do the task for you.
- Generate at least three virtual CVs and review the contents.

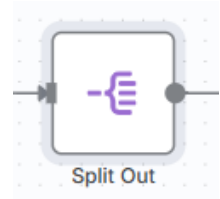
Example 2: Multiple CV Reading Machine

- Now we will improve our CV reading machine by reading multiple CVs at a time, which is our real daily life situation.
- You can place all your CVs into a new folder D:\CV\.
- Create a new worksheet in n8n and upload the flow *Multiple CV Reading Machine.json*:



Example 2: Multiple CV Reading Machine

- This flow will automatically read all the pdf inside the folder `d:\CV` and then feed into our AI Agent block one by one.
- After analysis, the results will be stored into separate files with the filename - *original filename_analysis.doc*.
- To achieve this, we use the new *Split out* node.
- It will split out the input files into separate files and feed into the AI Agent.



Split Out

Execute step

Parameters

Settings

Docs

Fields To Split Out

Use \$binary to split out the input item by binary data

Include

Options

Include Binary

☒

Add Field

Example 2: Multiple CV Reading Machine

- AI Agent outputs only the analysis result but no filename information. Therefore, when we store the results into new files according to our naming method, we need to modify the final *Read/Write File From Disk* node:

Read/Write Files from D... Execute step

Parameters Settings Docs

Use this node to read and write files on the same computer running n8n. To handle files between different computers please use other nodes (e.g. FTP, HTTP Request, AWS).

Operation

Write File to Disk

File Path and Name Fixed Expression

d:/CV/{{ \$('Split Out').item.binary.data.fileName.replace('.pdf', '_analysis.doc) }}_analysis.doc

Input Binary Field

data

The name of the input binary field containing the file to be written

Options

No properties

Add option

1. Select *Expression* in *File Path and Name*
2. Enter `"d:/CV/{{ $('Split Out').item.binary.data.fileName.replace('.pdf', '_analysis.doc' ) }}`

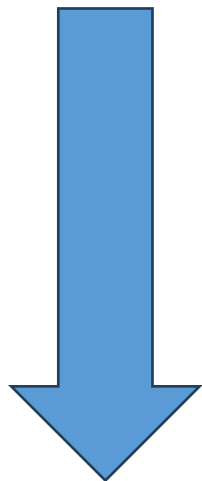
Note:

- It gets *fileName* from the *Split Out* node
- Replace the extension *.pdf* by *_analysis.doc*
- The result filename will be *fileName_analysis.doc*

Important Ground Concepts



The AI Evolution



Machine Learning

- Supervised and unsupervised learning
- Decision tree models
- Clustering....

Gen AI

- LLM (Large Language Model)
- Openai/ Gemini/ DeepSeek.....

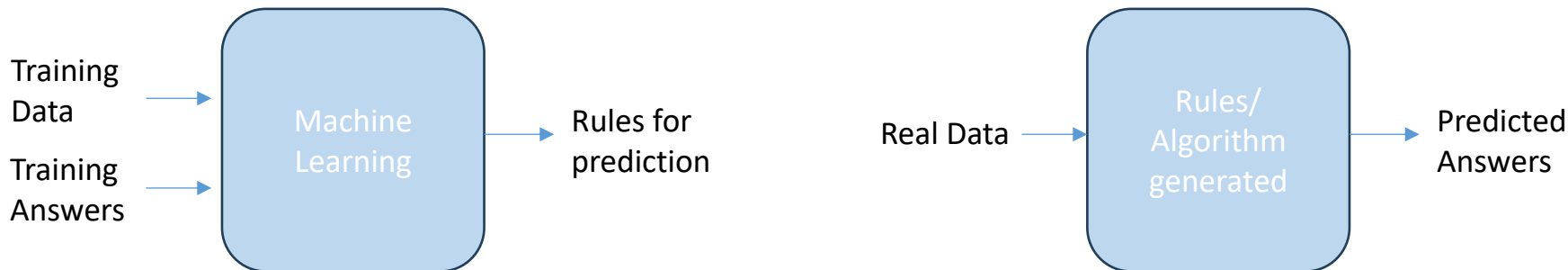
AI Agent

- Understand the context (reasoning)
- Enable the user of tools automatically

The AI Evolution – Machine Learning

Machine Learning

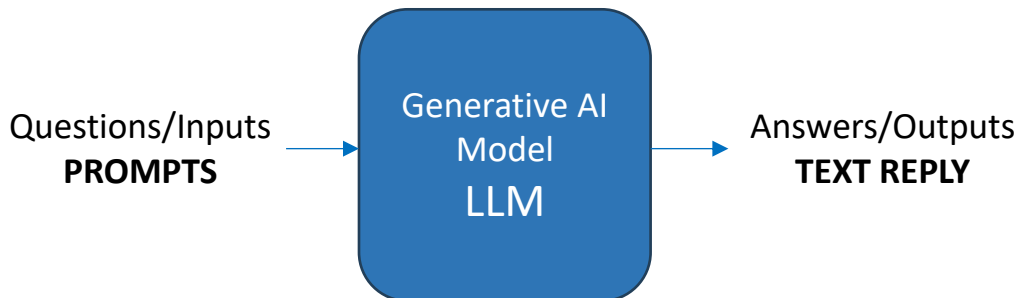
- Generate rules/algorithms from training data and answers provided.
- It will extract patterns from the training data and answers.
- Need specialists to work out



The AI Evolution – Generative AI

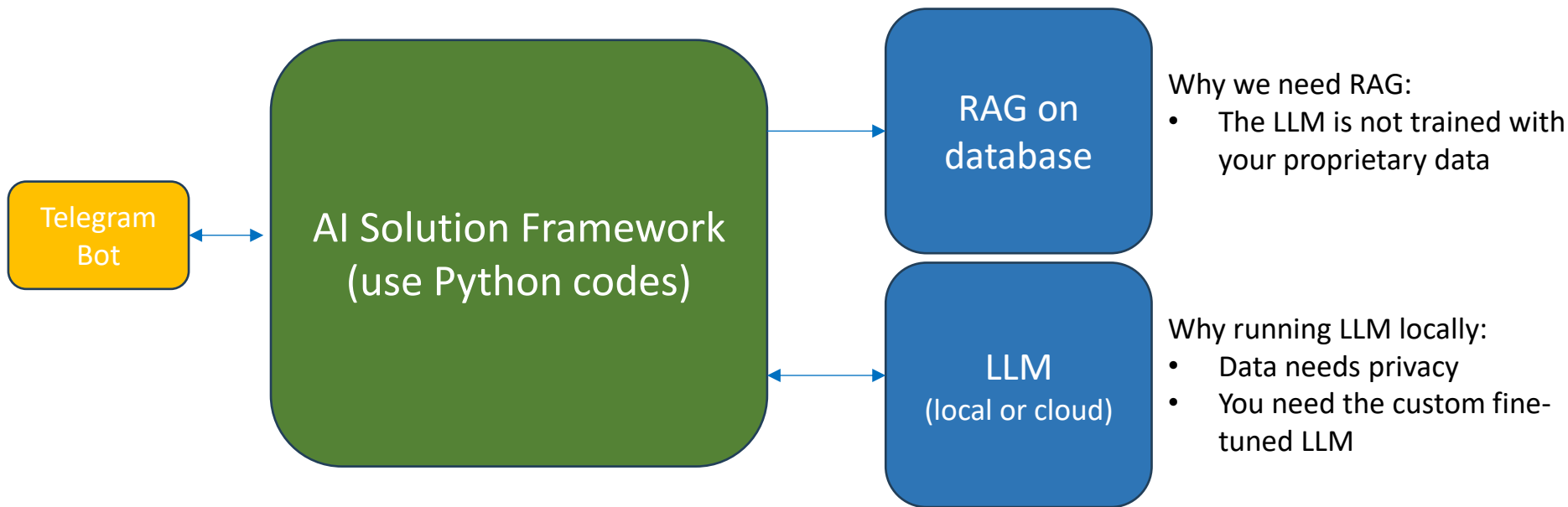
Generative AI

- LLM (Large Language Model) to understand human languages and generate context in human language.
- LLM is created via machine learning with a massive text dataset.
- Non-engineering people can use natural languages to apply AI.



More on AI Solution using LLM

Let's look at a typical block diagram of an AI ChatBot using LLM:

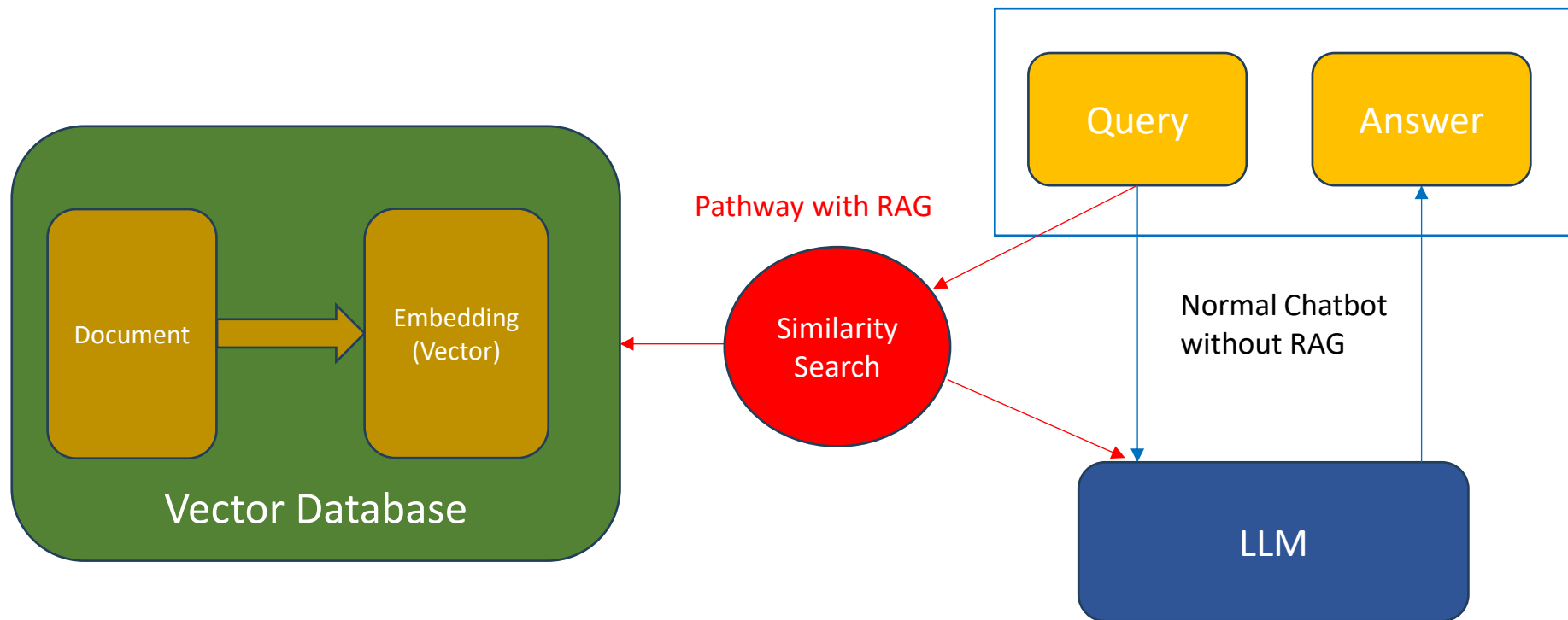


More on AI Solution using LLM

Fine-tuned LLM and Retrieval-Augmented Generation (RAG) can enhance the base LLM in different ways. Below summarize the differences:

Area	RAG	Fine-Tuned LLM
Core mechanism	Before injecting the prompt into the LLM, it will retrieve information from the external database to “enrich” this prompt based on the user query.	Modify the LLM permanently with the added dataset by training.
Cost & Resources	Lower upfront but higher latency per query	High cost for training but fast reaction
When to choose	Scalable, up-to-date responses like customer support over evolving documents	Need fast and consistent response

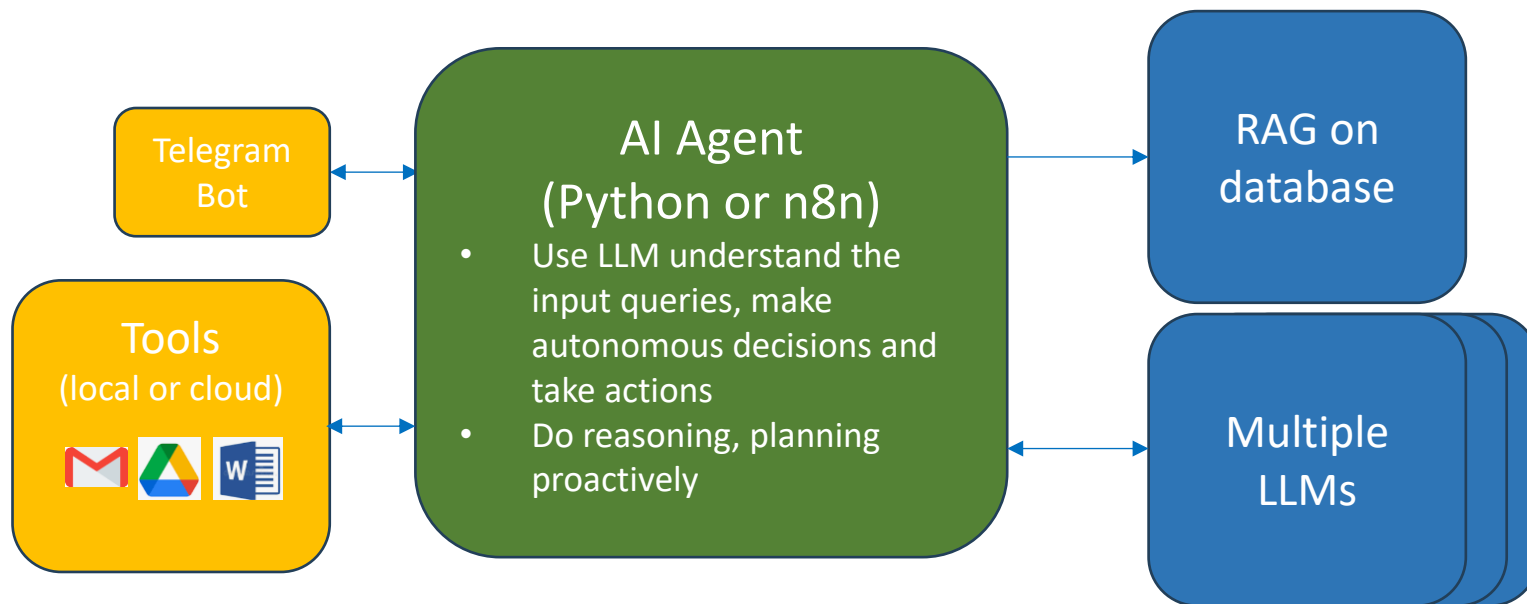
How RAG works



The AI Evolution – AI Agent

AI Agent

- Act like a human with his own brain and lots of tools (his hands)



THE END



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